

Diuretic

RII MYS-004
R.O.C. Reg. No.: 030521

Rasitol Injection 20mg/2mL



Acknowledgement

Draft

Ingredient(s):

Each ampoule (2 mL) contains:

Furosemide 20 mg
Benzyl Alcohol 0.02 mL

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamics:

1. Furosemide is one of the most potent diurectics. It generally produces an adequate increase in sodium and water excretion even in cases in which diuresis cannot be increased in other ways.
2. Furosemide inhibits reabsorption of sodium and water in the ascending limb of the loop of henle. The drug also decreases reabsorption of sodium and chloride and increases potassium excretion in the distal renal tubule and exerts a direct effect on electrolyte transport at the proximal tubule.
3. It is effective even in patients with considerably impaired glomerular filtration, hypoalbuminaemia or metabolic acidosis.
4. Furosemide lowers blood pressure initially by reducing plasma and extracellular fluid volume; cardiac output also decreases. Eventually, cardiac output returns to normal with an accompanying decrease in peripheral resistance.

Pharmacokinetics:

1. The half-life of furosemide is up to about 2 hours although it is prolonged in neonates and in patients with renal and hepatic insufficiency. It is up to 95% bound to plasma albumin, and is mainly excreted in the urine, largely unchanged. There is also some excretion via the bile and non-renal elimination in considerably increased in renal impairment.
2. Furosemide crosses the placental barrier and is distributed into breast milk. The clearance of furosemide is not increased by haemodialysis.

Indications:

Oedema due to cardiac, hepatic cirrhosis and nephritic syndrome. Treatment of mild to moderate degree of hypertension.

Dosage and Administrations:

For intravenous administration, furosemide must be injected or infused slowly; a rate of 4 mg per minute must not be exceeded.

For intramuscular administration, it must be restricted to exceptional cases where neither oral nor intravenous administration is feasible. Intramuscular injection is not suitable for the treatment of acute conditions such as pulmonary oedema.

Adult: For oedema, the usual initial dose is 1-2 ampoules once daily, adjusted as necessary according to response.

For hypertension, daily doses of 2-4 ampoules are given either alone or in conjunction with other antihypertensives.

Elderly: In the elderly furosemide is generally eliminated more slowly. Dosage should be titrated until the required response is achieved.

Rasitol Injection should be administered intramuscularly or intravenously.

To be used only by a physician.

Contraindications:

1. Furosemide should not be used in case of renal failure with reduced or absent urinary excretion (oliguria anuria) resulting from poisoning with nephrotoxic or hepatotoxic agents. This also applies to severely impair renal function with simultaneous liver failure and impaired consciousness (hepatic coma).
2. Severe hypokalaemia must be compensated before furosemide treatment. Potassium loss may be due to unbalanced diet, to vomiting or diarrhoea or to diseases of the liver or adrenal cortex and, in diabetics, to hyperglycaemia. In diabetics, a change of diet and medication may become necessary during furosemide treatment.
3. Severe electrolyte depletion (e.g. potassium or sodium) as well as serious reduction in blood volume (hypovolaemia with or without hypertension) and hypotension must be compensated before furosemide is administered.
4. Furosemide is contraindicated with patients who have a history of hypersensitivity to it as well as in women with childbearing potential.

Side Effect(s) / Adverse Reaction(s):

1. The most common side-effect associated with furosemide therapy is fluid and electrolyte imbalance including hyponatraemia, hypokalaemia and hypochloaemic alkalosis, particularly after large doses or prolonged administration. Because of its shorter duration of action, the risk of hypokalaemia may be less than thiazide diuretics.
2. Unlike thiazide diuretics, furosemide increases the urinary excretion of calcium. Nephrocalcinosis has been reported when furosemide has been used to treat preterm infants.
3. Furosemide may provoke hyperglycaemia and glycosuria, but probably to a lesser extent than the thiazide diuretics. It may cause hyperuricaemia and precipitate attacks of gout in some patients.

4. Signs of electrolyte imbalance include headache, hypotension, muscle cramps, dry mouth, thirst, weakness, lethargy, drowsiness, restlessness, oliguria, cardiac arrhythmias and gastro-intestinal disturbance.
5. Other side effects are relatively uncommon, and include blurred vision, yellow vision, dizziness, headache and orthostatic hypotension. Skin rashes, photosensitivity reaction and hypersensitivity reactions including interstitial nephritis, vasculitis, and fever occur rarely.
6. Pancreatitis is more common at high doses and cholestatic jaundice has been reported. Bone marrow depression may occur rarely: agranulocytosis, thrombocytopenia, and leucopenia have been reported.

Precaution(s) / Warning(s):

1. Although furosemide is used in high doses for oliguria due to chronic or acute renal insufficiency it should not be given in anuria or in renal failure due to nephrotoxic or hepatotoxic drugs nor in renal failure associated with hepatic coma.
2. Furosemide should not be given in pre-comatose states associated with hepatic cirrhosis.
3. It should be used with care in patients with prostatic hyperplasia or impairment of micturition since it can precipitate acute urinary retention.
4. Patient with manifest or latent diabetes mellitus should be checked regularly during furosemide treatment. The same applies to patients with gout.

Warning:

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age.

Not to be used in neonates.

Pregnancy and Lactation:

Furosemide should be administered in the first 3 months of pregnancy only if strictly indicated.

Also in the later stages of pregnancy, furosemide should be employed with circumspection.

Treatment during pregnancy requires monitoring of foetal growth.

Furosemide passes into breast milk and may inhibit lactation. Patient must stop breast feeding during the treatment period.

Interaction with others medicaments:

1. Furosemide may increase the ototoxic potential of aminoglycoside antibiotics, especially in the presence of impaired renal function. Except in life-threatening situations, avoid this combination.
2. Lithium generally should not be given with diuretics because they reduce lithium renal clearance and add a high risk of lithium toxicity.
3. Furosemide may add to or potentiate the therapeutic effect of other antihypertensive drugs. Potentiation occurs with ganglionic or peripheral adrenergic blocking drugs.
4. Furosemide may enhance the nephrotoxicity of cephalosporin antibacterials (e.g. cephalothin).
5. The diuretic effect of furosemide has been shown to be substantially reduced by concomitant antiepileptic therapy (i.e. carbamazepine and phenobarbitone).
6. NSAIDs may antagonise the diuretic effect of furosemide and other diuretics. Concomitant administration of NSAIDs with diuretics may increase the risk of nephrotoxicity.
7. Probenecid has been shown to reduce the renal clearance of furosemide and reduce the diuretic effect.
8. When a cardiac glycoside is administered concurrently, it should be remembered that potassium deficiency increases the sensitivity of the myocardium to digitalis.
9. Furosemide sometimes diminishes the potency of other drugs (e.g. the effect of antidiabetics and pressor amines) or potentiates their effect (e.g. in the case of salicylates, lithium and curare-mimetic muscle relaxants). The action of other hypotensive drugs may be potentiated by furosemide.

Symptoms and Treatment for Overdosage, and Antidote(s):

The principle signs and symptoms of overdose with furosemide are dehydration, blood volume reduction, hypotension, electrolyte imbalance, hypokalemia and hypochloremic alkalosis, and are extensions of its diuretic action.

Treatment of overdosage is supportive and consists of replacement of excessive fluid electrolyte losses. Serum electrolytes, carbon dioxide level and blood pressure should be determined frequently.

Shelf-life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature not exceeding 30°C. Protect from light. Avoid freezing.

Product Description and Packing(s):

A clear colourless solution.

2 mL x 100 Amps./Box



Manufactured by:
YUNG SHIN PHARMACEUTICAL IND. CO., LTD.
TAICHUNG YOUTH FACTORY
NO. 27, KUNG CHIU RD., TACHIA, TAICHUNG, TAIWAN, R.O.C.
Importer & Product Registration Holder:
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Date of revision: 25 Nov 2019